



Friction and wear behaviour of additively manufactured continuous carbon fibre reinforced PA6 composites

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ABSTRACT

In the present work, continuous carbon fibre (CCF) reinforced polyamide 6 (PA6) composites were fabricated using the Fused Filament Fabrication (FFF) 3D printing technique. The sliding friction and wear behaviour of the printed CCF/PA6 composites was investigated as a function of varying fibre orientations (i.e. normal, parallel, and anti-parallel) and layer deposition directions (i.e. parallel and perpendicular) with respect to the sliding direction. It emerged that fibre orientation played a key role in determining the wear resistance of 3D printed composites. Severe fibre breakage was observed when the orientation of fibre was in normal direction against the sliding direction, associated with a higher wear rate than that obtained in parallel or anti-parallel direction. To better understand the effects of voids on wear mechanism, tribo-tests were further carried out with the specimens that underwent compression moulding (CM) post-treatment process. The results confirmed that the internal defects could cause adverse impacts on wear performance, depending on the fibre orientations. In particular, it was found that fibre distribution, as well as the fibre/matrix interface within the filament, was the crucial parameter affecting the overall tribological performance of the printed structures. The findings add to our understanding of designing and fabricating wear-resistant polymer composites using 3D printing technologies.

1. Introduction

Polymers and polymer composites are extensively used in various tribological applications such as rollers, gears, and dry sliding bearings, thanks to their intrinsic advantages such as high specific strength, corrosion resistance and self-lubricating behaviour [1,2]. It is widely accepted that the tribological performance of polymeric materials can be improved by incorporating various fillers or reinforcing fibres [3–5]. In particular, compared with short fibres, continuous fibre reinforcements allow a higher fibre volume fraction and more effective load-transfer between matrix and fibres, and thus the reinforcements could effectively enhance the load-bearing capacity and wear resistance of polymer composites [2]. However, conventional manufacturing techniques (e.g., resin transfer moulding, automated tape laying, automated fibre placement, filament winding, and pultrusion etc.) employed to fabricate continuous fibre reinforced polymer (FRP) composites are often limited by expensive moulds and metal tools, as well as restricted design flexibility [6,7]. Alternatively, with continuing development and wider adoption of 3D printing (also referred to as additive manufacturing, AM)

technology, the fabrication of additively manufactured continuous FRP composites has gained increasing attention in recent years. This is mainly attributed to its ability to transform complex geometries to the final shape without the need of moulds or special tools. There are three major approaches to print continuous FRP composites: i) in-situ fusion of thermoplastic liquid resin before extrusion [8]; ii) in-situ fusion of molten thermoplastic with fibres at the nozzle [9,10] and iii) extrusion of pre-impregnated fibres [11–14]. The last approach together with FFF becomes the most commonly used additive process for manufacturing continuous FRP composite parts [15,16].

Up to now, studies on additively manufactured polymer composites are carried out mainly to improve the basic mechanical properties such as strength and fracture toughness, by: a) analysing the effect of process parameters on the mechanical behaviour of specimens [11,17]; b) the failure modes to better understand the microstructural failure mechanisms [18,19] and c) developing new fibre-matrix filaments and novel printing techniques [20,21]. Little has yet been focused, however, on their wear and friction properties. In the field of tribology, pioneering work has been conducted on particle and short fibre reinforced polymer

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composites. Dawoud et al. [22] studied the tribological properties of the additively manufactured polymers with additional reinforcements, e.g., by incorporating graphite flakes in Acrylonitrile Butadiene Styrene (ABS) matrix. Despite being a well-known solid lubricant, the addition of fillers resulted in a significant increase in both friction coefficient and wear rate. The phenomenon might be related to the poor interface bonding between the fillers and polymer matrix, which is a common issue for the printed composite materials. Therefore, it was suggested that fabrication parameters and fillers selection should be further investigated. Indeed, a number of studies have demonstrated that the mechanical properties and wear behaviour of printed parts could be largely affected by the processing parameters, such as raster angle, layer thickness and printing orientation [23–26]. More recently, Singh et al. [27] developed a PA6 based filament with hybrid reinforcements, namely, SiC and Al₂O₃. An empirical model was proposed to predict the wear properties of printed samples, showing the potential of designing the composite materials with desirable properties using 3D printing technology. Besides the traditional micro-sized fillers, the beneficial effects of nanofillers on the printed composites were also reported. For instance, Lin et al. [28] printed short carbon fibre reinforced polyetheretherketone (PEEK), coating on pure PEEK backings. It was found that the friction performance of the coating material can be effectively improved with the induced nanosilica particles. Jeon et al. [29] successfully improved the wear-resistance of 3D printed thermoplastic elastomer with the addition of surface-modified carbon black and electrochemically exfoliated graphene fillers. In particular, thanks to the good dispersibility and lubricating effect of graphene, the mechanical properties and wear resistance of the composites can be effectively improved even with 1 wt% nanofillers. Finally, it is worth mentioning the novel additive manufacturing-based recycling approach presented by Liu et al. [30] who recently processed FRP waste into 3D composites parts with the desirable mechanical and tribological properties. The work illustrated the possibility to use 3D printing method for recycling polymeric wastes and thus improving environmental sustainability. The aforementioned work has well demonstrated that AM technologies are promising for fabricating tribo-parts with complex geometries and further tailored tribological performance by incorporating various fillers, to meet challenging design requirements from modern industries. Nevertheless, until now, little research has been done to investigate the friction and wear behaviour of additively manufactured continuous FRP composites.

In the present work, unidirectional continuous carbon fibre reinforced PA6 composites have been additively manufactured using the FFF technique. The friction and wear behaviour of the composites is studied using a pin-on-disc apparatus under dry sliding conditions. The wear mechanisms are investigated with the assistance of scanning electron microscopy (SEM) observations. The work has focused on evaluating the tribo-effects of internal structures of the printed composites, such as fibre orientations, layer deposition directions and the internal voids. In addition, the compression moulded composite specimens are fabricated to serve as a comparison with the ones manufactured by the FFF method. The study aims to provide an in-depth understanding of the basic processing-structure-property relationship of 3D printed continuous FRP composites as the high wear-resistant materials for tribological applications.

2. Experimental

2.1. Materials and sample fabrication

In this study, a commercially available Mark Two desktop 3D printer (Markforged) was used to 3D print unidirectional CCF/PA6 plates with the layer height pre-set at 0.125 mm. The raw material used was a 0.4 mm diameter pre-impregnated continuous carbon fibre reinforced PA6 filament with approximately 35% fibre volume fraction (CFF, Markforged, USA). The samples were printed with the speed of ~15 mm/s,

and the printing temperature of ~275 °C. In addition, the neat PA6 samples were also printed under the same conditions for comparative analysis.

The cross-section of raw CCF filaments was observed using an optical microscope, as shown in Fig. 1a. It is noted that the distribution of continuous carbon fibres within a filament was rather uneven. The dark regions between adjacent fibres indicated the existence of microscopic voids. Fig. 1b shows the cross-section of the printed sample, where the interface between layers can be identified.

To further quantify the void content of printed CF/PA6, representative samples (4 × 4 × 6 mm) were scanned using an X-ray scanning system (Bruker SkyScan 1272) at 30 kV and 150 μA with the pixel resolution of 3.25 μm. Substantial voids can be observed from Fig. 1c. It can be seen that voids mostly exist between adjacent contacting filaments/layers. By further reconstruction and analysis of the scanned specimens, the void content can be determined as ~10% for the bulk sample. To investigate the influence of voids, the compression moulding (CM) process was introduced after the printing process. As-printed specimens were placed in a heating mould and kept at 230°C. Then, 5 MPa pressure was applied and held for 10 min.

2.2. Sliding wear tests

Unlubricated sliding wear tests were conducted at room temperature via a pin-on-disc configuration by using a NANOVEA-MT/60/NI tribometer, as shown in Fig. 2. The sample pins for wear tests were in the dimensions of 4(L) × 4(W) × 10(H) mm, which were cut from large-sized 3D printed cuboids (20 × 40 × 150 mm), followed by a cleaning and drying process before wear tests. Different positions were selected to test in order to assure the universality and accuracy of the results. The steel counterparts were commercially available bearing washers (SKF-LS2542) with the hardness of approximately 880 HV and surface roughness, *Ra*, of 0.25 μm. The load-carrying capacity of the composites was evaluated by applying different normal pressures of 0.5 MPa (8 N), 1 MPa (16 N) and 2 MPa (32 N), whereas the linear sliding speed was kept at a constant value of 0.5 m/s. The testing time was 20 h, and at least three repeated tests were performed for each condition to ensure the reliability of the data. During the sliding process, the coefficient of friction, *μ*, was collected by the tribometer at a frequency of 0.1 Hz. Then, the mean values of the friction coefficient during the steady-state of sliding process were determined as the friction coefficient of the sample for further comparisons. After the test, the mass loss was measured by differentiating the weight of the specimen before and after the wear test. The specific wear rate, *W_s*, was then determined by Equation (1):

$$W_s = \frac{\Delta m}{\rho \times F_N \times L} \left(\frac{\text{mm}^3}{\text{Nm}} \right) \quad (1)$$

Where *Δm* is the mass loss of composite pin, *ρ* is the density of the specimen, *F_N* is the applied normal force, *L* is the total sliding distance. The density of composite samples was measured in accordance with ASTM D792 on Sartorius® weighing balance with an accuracy of 0.00001g, as per Equation (2):

$$\rho_c = \rho_{\text{liquid}} \times \frac{W_{\text{air}}}{W_{\text{air}} - W_{\text{liquid}}} \quad (2)$$

where *ρ_c* is the density of composite, *ρ_{liquid}* is the density of still water used as a medium (1g/cm³), *W_{air}* is the weight of a sample in air, *W_{liquid}* is the weight of a sample fully immersed in liquid. In order to evaluate the durability of materials under different loading conditions, the time-related depth wear rate, *W_t*, was introduced, as per Equation (3):

$$W_t = k^* \cdot p v = \frac{\Delta h}{t} \text{ (m/s)} \quad (3)$$

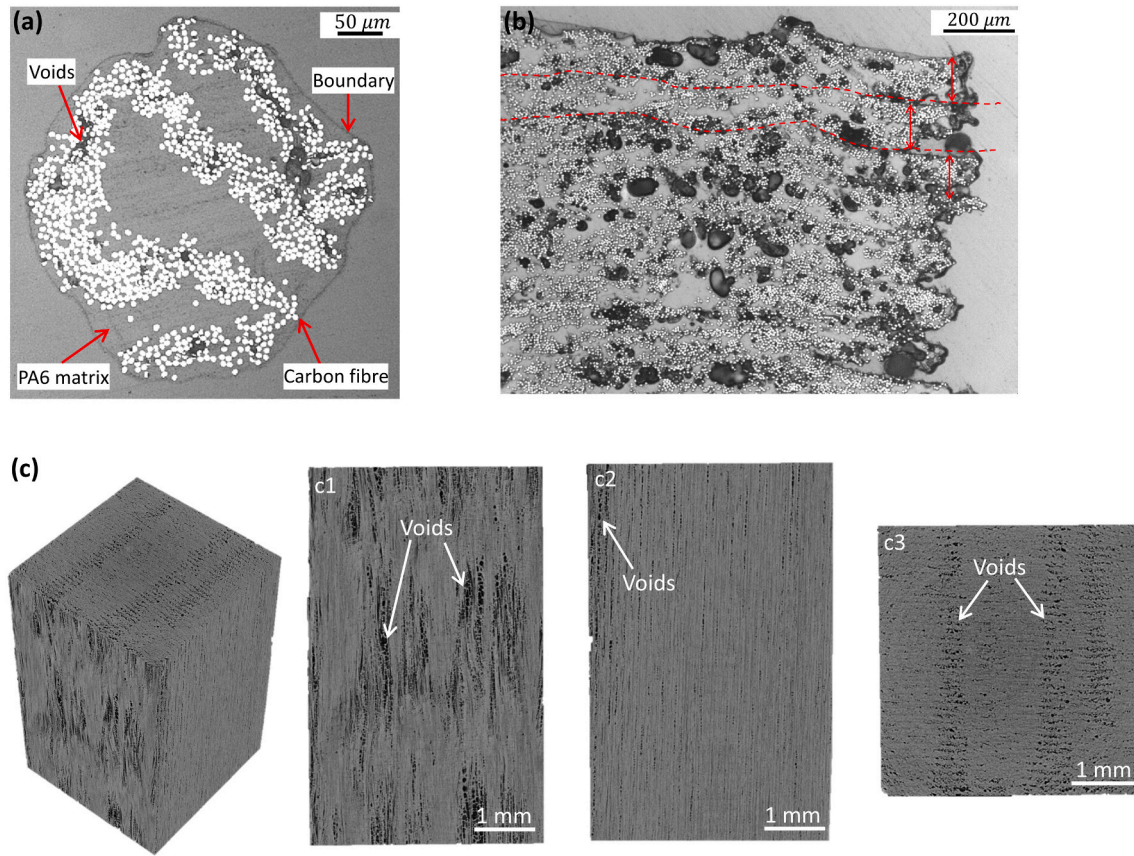


Fig. 1. Micrographs of the cross-sections of (a) a raw filament, (b) the printed sample (the inserted arrows show the deposited layers, and the dash lines show the layer interface), and (c) typical morphology of 3D printed CCF/PA6 components by micro-CT.

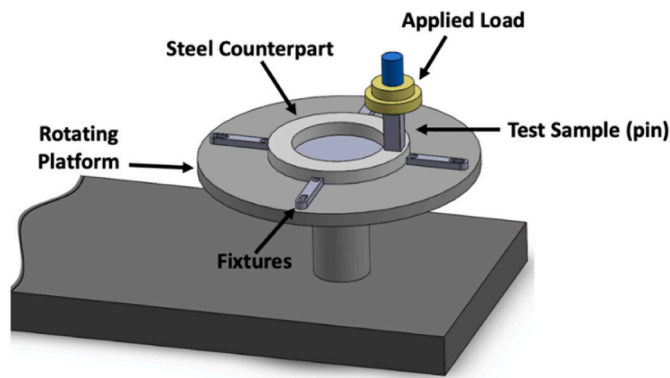


Fig. 2. Schematic illustration of sliding wear test.

where k^* is the basic wear factor of the material, p is the normal pressure, v is the sliding velocity, Δh is the height reduction of the specimen, and t is the test time.

The wear tests were designed and conducted, considering three fibre orientations - parallel (P), anti-parallel (AP) and normal (N), in relation to the sliding direction of the counter body, as depicted in Fig. 3a. In addition, when fibres were normally oriented to the counterpart, two printing layer orientations were considered, i.e. perpendicular and parallel in the composites with respect to sliding direction, as shown in Fig. 3b. After the wear test, the surface profiles of both polymer pins and steel counterparts were examined by using an Olympus 3D laser scanning microscope. SEM Zeiss Sigma HD was used to investigate the morphologies of worn surfaces with 5 kV accelerating voltage. Before SEM observation, the worn surfaces were coated with ~10 nm sputter

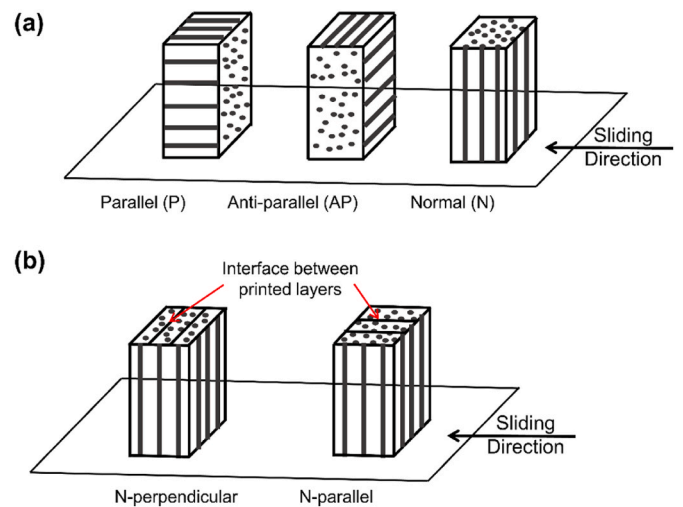


Fig. 3. Sliding directions during the wear tests: (a) Fibre orientations: parallel (P), anti-parallel (AP) and Normal (N); (b) Printing layer orientations with fibres in normal direction: N-perpendicular and N-parallel.

gold.

3. Results and discussion

3.1. Friction and wear results

Fig. 4 shows the typical friction and wear results for neat PA6 tested under different loads. Under the relatively low load, i.e. 0.5 MPa, the

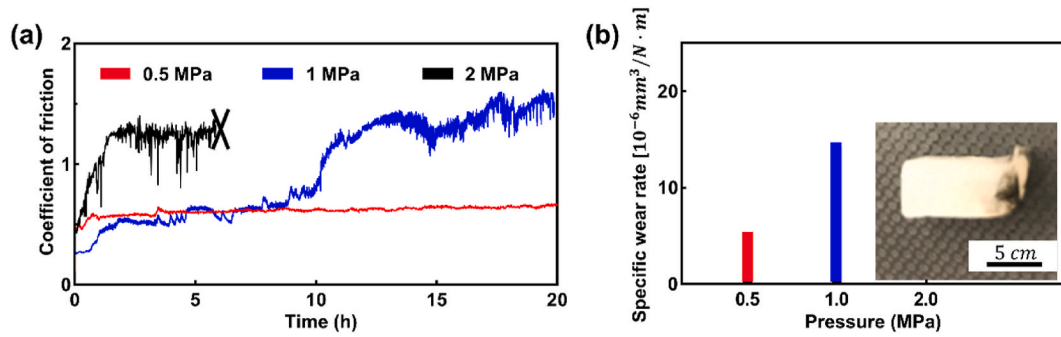


Fig. 4. (a) The friction and (b) wear behaviour of neat PA6 tested under different loads.

neat matrix showed stable friction and wear performance. Under 1 MPa, however, the friction coefficient sharply increased after 10 h, with a high value of ~ 1.5 with larger fluctuation, which can be explained by friction heating due to the strong adhesion between the PA6 sample and the counterpart covered with transferred PA6 debris. Accordingly, the wear rate was greatly increased. With the further increase of normal load up to 2 MPa, neat PA6 failed after ~ 6 h with significant deformation on a macro scale because of the severe thermal softening (see the inserted picture in Fig. 4b). In this case, the wear rate was not measured.

With fibre reinforcements, the load carry capacity of CCF/PA6 composites can be effectively enhanced. Table 1 summarises the friction and wear results for the printed CCF/PA6 composites tested under different sliding conditions. All the composites could survive under 2 MPa, with much less wear loss compared to those obtained from neat PA6. It is also noted that the tribological performance of CCF/PA composites is greatly dependent on the fibre orientations against the sliding direction.

Fig. 5 compares the representative friction processes as functions of sliding time under different sliding directions. It is noticed that for the samples in P- and AP- orientations, the coefficient of friction increased at first, followed by a decrease to a relatively stable value. The initial increase in friction may be due to the polishing effect on the as-printed specimens, resulting in a higher adhesive force with the enlarged real contact area. On the other hand, during the running-in stage, transfer films would develop on steel discs consisting of wear debris of polymer matrix, small broken fibre elements and sometimes wear powders of counterpart asperities [31,32]. The friction became steady when the surface properties of sliding pairs were stable. In particular, with the formation of transfer film, the dynamic balance would be achieved in the steady stage, i.e. the removed transfer film during wear process would

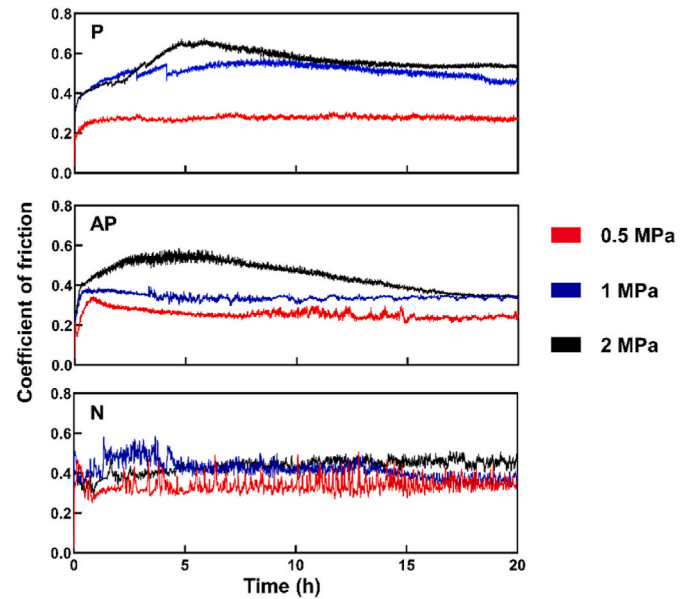


Fig. 5. Representative evolution of the friction coefficients in the P-, AP-, and N-fibre orientation as a function of sliding time.

be immediately replenished by new wear debris [33,34]. However, when the sample slid in N-direction, there was no noticeable running-in process, and friction coefficients showed more fluctuations, suggesting the different contact mechanisms from P- and AP- directions.

Table 1

Tribological properties of CCF/PA6 composites under various wear conditions.

Sliding direction	Pressure (MPa)	Frictional coefficient	Specific wear rate ($10^{-6} \text{ mm}^3/\text{Nm}$)
Parallel	0.5	0.37 ± 0.12	0.42 ± 0.32
	1	0.49 ± 0.1	2.08 ± 1.41
	2	0.53 ± 0.11	2.31 ± 1.09
Anti-parallel	0.5	0.31 ± 0.05	0.66 ± 0.58
	1	0.40 ± 0.07	2.34 ± 1.27
	2	0.43 ± 0.09	2.45 ± 1.34
Normal	0.5	0.36 ± 0.07	3.26 ± 0.46
	1	0.40 ± 0.05	5.54 ± 1.38
	2	0.45 ± 0.05	12.56 ± 2.45
N-perpendicular	0.5	0.38 ± 0.06	3.45 ± 0.42
	1	0.42 ± 0.03	5.47 ± 0.19
	2	0.45 ± 0.06	13.57 ± 2.11
N-parallel	0.5	0.35 ± 0.09	3.07 ± 0.48
	1	0.39 ± 0.06	5.95 ± 2.03
	2	0.44 ± 0.05	11.56 ± 2.75

Fig. 6 summarises the tribological results from different testing conditions. It can be seen that friction and wear tend to increase with ascending normal loads, and a similar tendency was also observed for the conventionally manufactured polymer composites [35,36]. In general, the more severe sliding conditions, i.e. high load pressure and/or sliding speed, would aggravate the wear process due to the accelerated thermal-mechanical failure caused by the high stress and/or contact temperature. It is worthwhile noting that compared with P- and AP-orientations, the carbon fibre composites tested in N orientation underwent the highest wear rate, associated with different characteristics of friction process (cf. Fig. 5). Despite the similar trends, there is also a difference between P- and AP- orientations. In particular, the friction force in P- orientation is higher than that in AP- orientation, especially when the load is high. The similar phenomenon was also noticed by Zhang et al. [37], which might be related to the anisotropic properties of carbon fibre. In general, the mechanical properties along the fibre axis are much stronger than perpendicular to the axis. On the other hand, corresponding to the different fibre orientations, composites may show different failure behaviour with fibres [38]. All those factors may contribute to the different friction forces in different orientations. Such an orientation-dependent tribological behaviour has also been widely reported and studied with conventionally manufactured polymer composites [39–41]. To understand the underlying wear mechanisms, the worn surfaces of different samples were further investigated using SEM, which will be discussed in the following sections.

3.2. Effects of fibre orientations

3.2.1. Microscopy analysis of worn surface

Fig. 7 shows the worn surfaces of CCF/PA6 composite pins tested in P- and AP- orientations subjected to different loading conditions. Under a relatively low applied pressure, both specimens in P- and AP- orientations showed smooth worn surfaces. Mild fibre/matrix debonding and breakage of fibres were observed, as shown in Fig. 7a and b. With the increase of the load, the fibre/matrix interfacial debonding progressively worsened. Consequently, fibres suffered more damages without protection from the surrounding matrix (Fig. 7c and d). This phenomenon was more significant in AP direction, as the matrix between near fibres in AP direction was more likely to be removed due to high shear stress induced by fibres. Thus, fibre/matrix debonding and fibre breakage was the dominant factor in the wear of AP orientation [38]. Also, some loose fibres were pulverised or crushed into smaller pieces, and the fibre fragments and particles could either be lost from the contact area or remain in contact regions (Fig. 7d). When the applied pressure was further increased, as shown in Fig. 7e and f, more fibre breakage and removals occurred, accompanied with being compacted on the surface. However, it was also noticed that the wear debris of the PA matrix could be deformed and transferred back on the worn surface. Those debris somewhat repaired the fibre/matrix interface by filling the cracks/cavities on worn surfaces. This may explain the results that the

specimens showed relatively stable specific wear rates when the load pressure increased from 1 MPa to 2 MPa (cf. Table 1).

Fig. 8 shows the worn surfaces of composite pins in N- orientation, which is distinctly different from those resulted from P- and AP- directions. As shown in Fig. 8a, wear debris powders loosely attached on the surface, which were normally generated under abrasive wear process. In normal direction, fibre nodes protruded against the metal counterface. Due to stress concentration, damages at fibre tips such as breakage and pulverisations occurred, especially under relatively high load pressures. By comparing Fig. 8b and c, deeper furrows on worn surfaces are also observed, suggesting abrasion as the dominant wear mechanism. The severe abrasive wear with more fibre damages may explain the high wear rates shown in Fig. 6, and fluctuations of frictions in N direction illustrated in Fig. 5.

To further understand the evolution of fibre failure, as well as the governing wear/contact mechanisms in 3D printed composites during the wear process, the worn surfaces in normal direction were traced at various sliding durations, i.e. 1h, 3h, 5h, and 10h, respectively. During the first 1 h, as shown in Fig. 9a, fibre cracking (A) and matrix push-ups (B) were observed, accompanied by the initiation of fibre/matrix debonding (C). With the increase in sliding distance, crack propagation occurred along the fibre/matrix interface, and more noticeable fibre damages were presented (Fig. 9b, A). After 5 h, fibre pulverisation could be seen at the tips of fibres (Fig. 9c, A). As wear test continued, associated with severe debonding failure at fibre/matrix interface, fibres were fractured and spalled from the matrix, resulting in obvious cracks and cavities (Fig. 9d). Some of the large-sized fibre fragments were noticed in the contact region between the specimen and counterpart. Those rigid fibre debris would act as the abrasive mediums between sliding pairs, resulting in three-body abrasive wear. This also explains the micro-sized grooves (Fig. 9d, A) and high wear rate in N- direction, especially under high loading conditions.

3.2.2. Examination of surface profiles

After the wear tests, the surface profilometry measurements were conducted on the sliding pairs, including metallic discs and polymer pins. When polymer specimens slide against the metallic counterparts, it is well known that transfer films can develop on the counterparts and largely determine the sliding behaviour of the sliding system [42,43]. Fig. 10 shows the worn surfaces of steel counterparts after the wear tests against composites in different fibre orientations. As shown in the figures, the counterfaces rubbed against P and AP samples were covered by continuous, thin transfer films, with the increased smoothness of the surface comparing to the virgin steel samples. In contrast, deep grooves were observed on the steel disc of N specimens without the continuous transfer films. The grooves in sliding direction indicated that the wear process was governed by the two-body and three-body abrasive wear. The observation corresponds to the worn surface of composite pins (cf. Figs. 7 and 8). Transfer films developed when fibres in-plane alignment, contributing to a rather stable specific wear rate. In N- orientation,

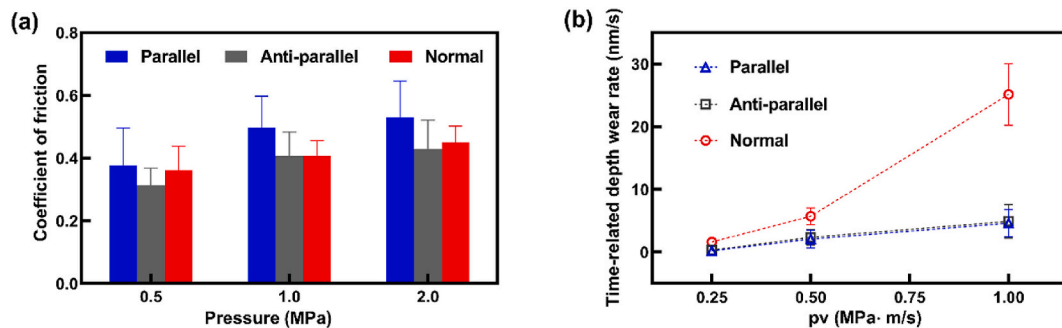


Fig. 6. Tribological behaviour among three fibre orientations: (a) coefficients of friction as a function of nominal pressure, and (b) time-related depth wear rate as a function of pv factor.

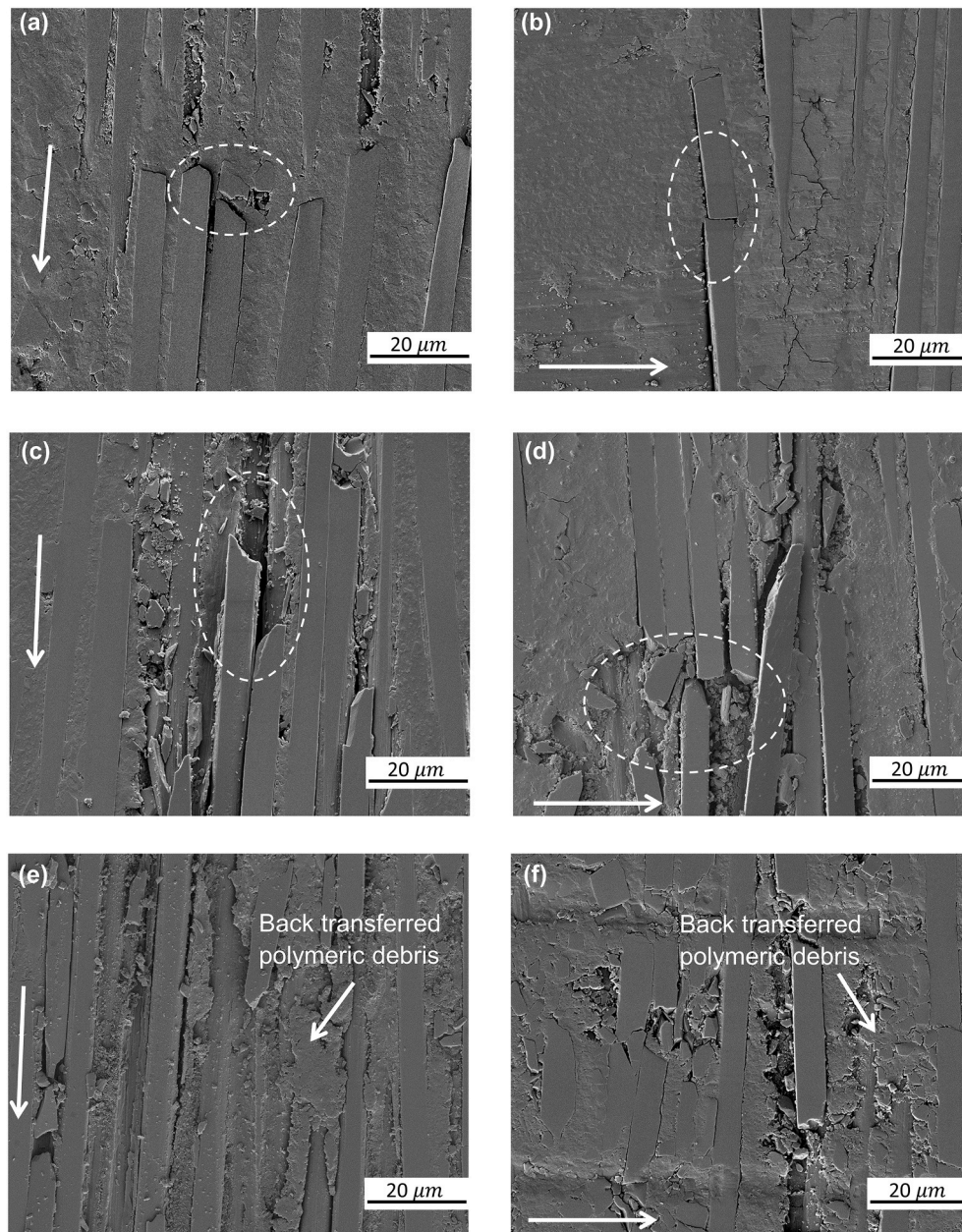


Fig. 7. Worn surfaces of specimens in P and AP fibre orientation under different loadings: (a, b) 0.5 MPa; (c, d) 1 MPa; (e, f) 2 MPa at high magnification. The arrows indicate the sliding direction.

however, the standout fibres and large-sized fibre debris acted as abrasive medium scratching the tribo-parts resulting in deep furrows along the sliding direction on the worn surfaces of sliding pairs.

As shown in Fig. 11, for P- orientation, the surface roughness of the worn polymer pin matched that of the steel disc. Also, “back-transferred” polymer debris can be seen (which were also observed in Fig. 7e and f). The results confirmed the adhesion wear mechanism with the presence of the polymeric transfer film at the interface between the sliding pair. However, for N-orientation, the worn surface of the polymer pin was much smoother than that of the steel disk. This can be explained by the three-body abrasive wear caused by wear debris of hard fibre particles. As shown in figures, fibre bundles can be identified. As aforementioned, the tips of fibre can be crashed, resulting in fibre particles. Those particles can be subsequently trapped in the bundle zones of fibre (Fig. 8b). Without the formation of effective transfer film layers, the hard fibre debris can cause severe three-body abrasive on steel disks. It is worthwhile noting that the size of grooves on steel disks ranges from

tens to a few hundreds of micrometres, which are comparable to the exposed fibre bundles shown in Figs. 1b and 8.

3.2.3. Discussions on wear mechanisms

From the above morphologies, it is clear that for all fibre orientations, the matrix materials were quickly worn away, and the rigid fibres would be exposed to counterpart surfaces, carrying most of the normal load. Thus, the wear performance of polymer composites is governed by the wear mechanisms depending on the failure/removal process of fibre reinforcements. When the fibre axis was in the plane of contact (i.e. P- or AP-orientation), partially-worn fibres protruded slightly above the matrix, contributing to the real contact areas, which would be much larger than those created by fibres in N-orientation. Accordingly, fibres in-plane contact would undertake less damage than those in normal direction. Fibre thinning was the predominant removal process of fibres in P and AP directions. This would result in sheet-shaped fibre debris, which is beneficial for the formation of transfer film on the steel

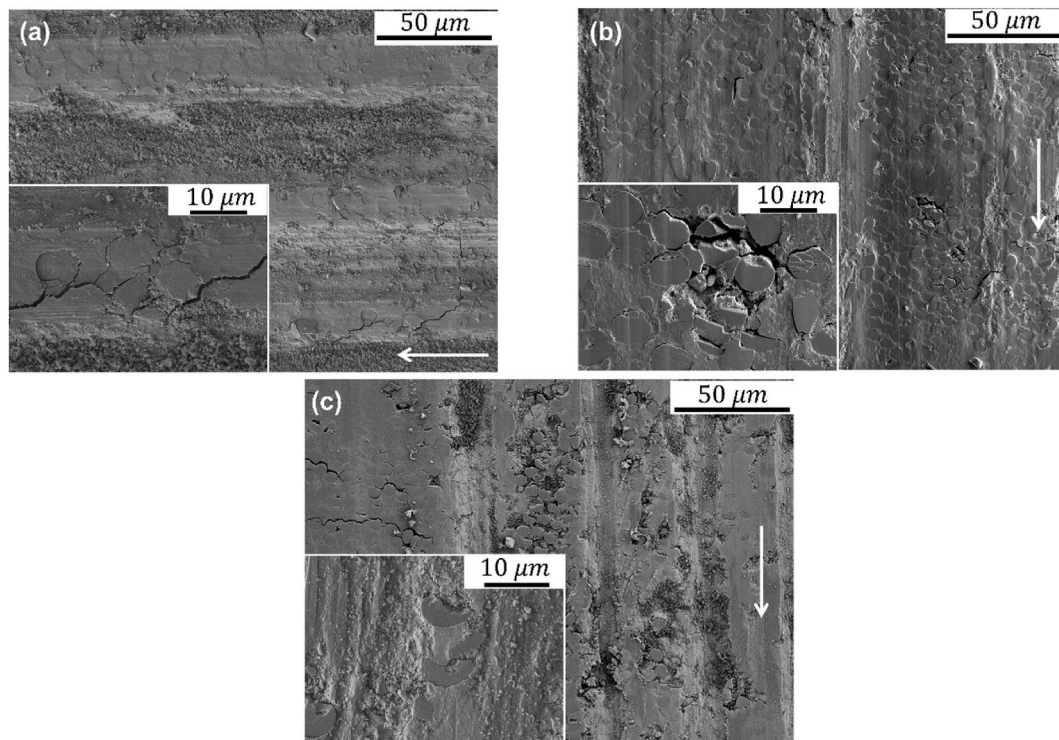


Fig. 8. Worn surfaces of specimens in N fibre orientation under different loadings: (a) 0.5 MPa; (b) 1 MPa; (c) 2 MPa. The arrows indicate the sliding direction.

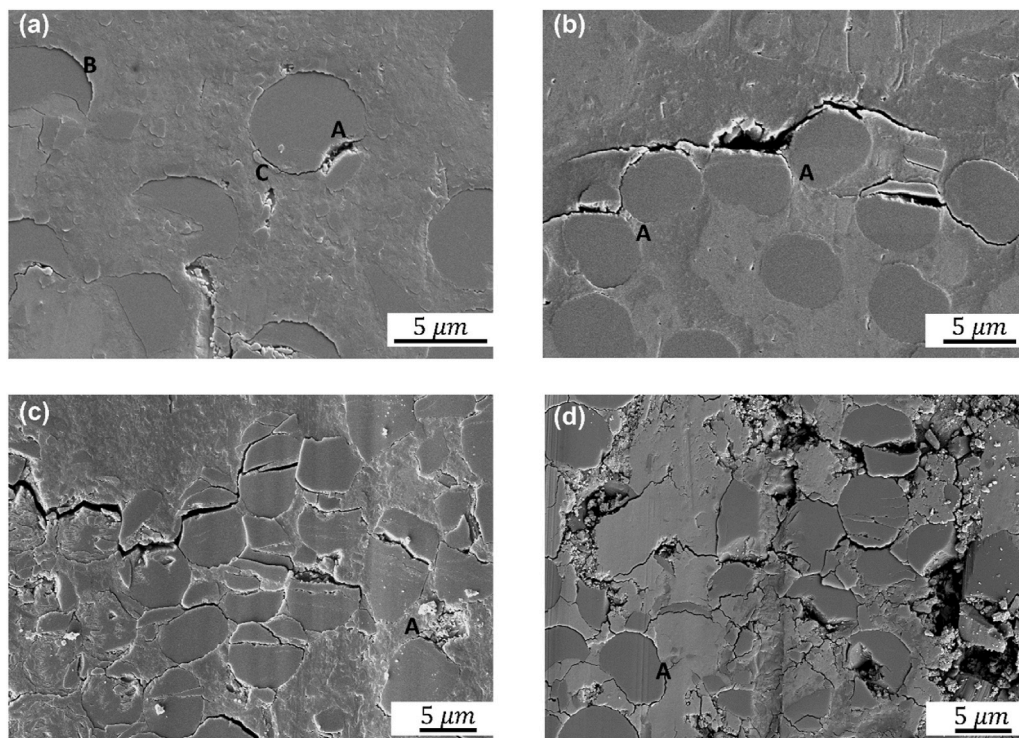


Fig. 9. Worn surfaces of specimens in N fibre orientation with different testing durations: (a) 1h; (b) 3h; (c) 5h; (d) 10h.

counterface [44,45]. In the case of N orientation, stress was concentrated on tips of protruding fibres, leading to an extensive level of fibre damage such as fibre breakage and pulverisation. Particulate fibre debris were noticed in the contact region, which might contribute to the three-body abrasive wear rather than the formation of transfer film. SEM observation confirmed that the wear process was dominated by abrasion

wear with deep grooves/furrows along the sliding direction.

It is worth mentioning that fibre/matrix separation generally occurred despite fibre orientations, which indicates that fibre/matrix interfacial bonding is relatively weak for additively manufactured composites. However, with the poor interfacial bonding, the upper load limit of the fibres tends to reach under relatively low loading when tests

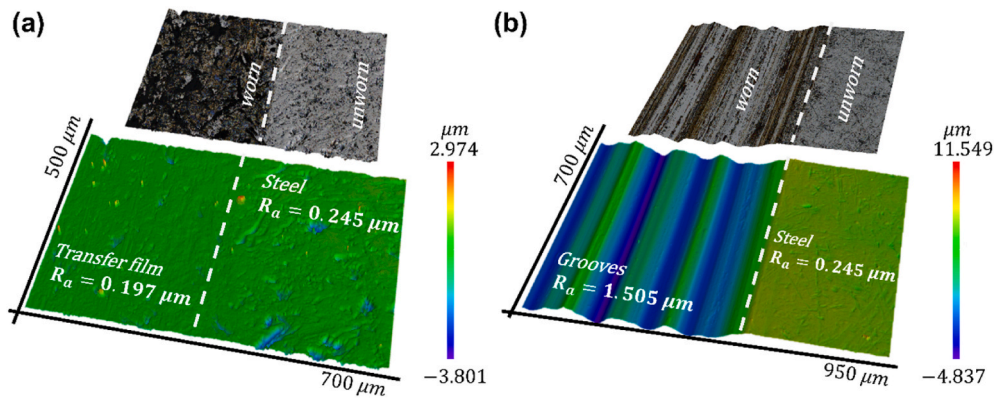


Fig. 10. Surface profiles of the steel counterfaces which have been slid against: (a) P-; (b) N-fibre orientated polymer composite samples.

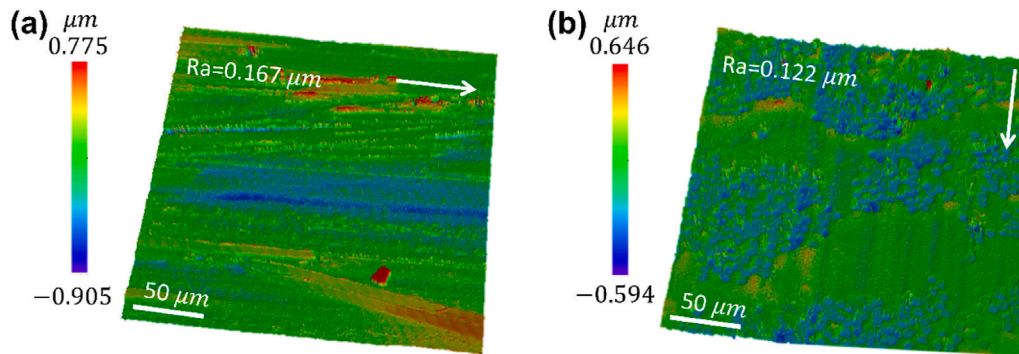


Fig. 11. Surface profiles of the worn polymeric pins sliding in: (a) P-; (b) N-fibre orientation.

were conducted in N-orientation [40,41]. This agrees with the wear results shown in Fig. 6. The results could also be verified and explained by a finite element simulation of stress conditions when unidirectional composite subjected to a moving asperity contact under various fibre orientations [46]. In particular, for P and AP orientation, fibre/matrix debonding may take place either at the surface region or underneath the direct contact area down to a depth of one to two fibre diameters, depending on the applied load. During wear process, fibre units are grounded gradually, or sometimes broken into larger pieces, which remain in the original places and contribute to the subsequent wear process (cf. Fig. 7). It is worth noting that, although marginally higher, the difference in wear rates between transverse (AP) and longitudinal (P) direction could be ascribed to the more occurrence of debonding in the AP case [41,47]. On the contrary, under N orientation, fibre/matrix debonding will take place at the surface and propagate down along fibre axis to a finite distance. The length of debonding will not change until the fibre length is decreased by wear process [39]. It was indicated by FEA studies that when fibre reinforced composites being slid in N orientation, there existed tensile stress on the surface behind the contacting asperities and compressive stress before the asperities, and such repeated compression-tension loadings would lead to fatigue wear and breakage of fibres [38,48–50]. Subsequently, fibre might be fractured next due either to buckling or to a mechanical pulling at the surface. Strong interfacial bonding makes such phenomenon difficult to happen, as the deformation of fibres will be constrained by the presence of neighbouring matrix and fibres. However, poor fibre/matrix interfaced bonding would lead to the failure of the matrix in interface regions (Fig. 9d). Thus, due to the lack of surrounding restraints, fibre breakage and removal could occur, which in turn results in a deeper debonding zone. Accordingly, once the support from intact fibres reduces, the matrix will be subjected to more micro-ploughing and micro-cutting attacks by asperities on the contact area, contributing to more wear loss.

3.3. Effects of layer deposition direction

Fig. 12 presents the tribological performance of composites when sliding takes place in N direction of fibres, but relative to different printing layer directions, i.e. parallel and perpendicular. According to the results, the friction properties of the studied tribo-pair were insensitive to printing layer orientation. By comparing results in Fig. 12b, there was a slight difference in wear resistance under the relatively high applied pressure of 2 MPa. Nevertheless, it can be seen that wear behaviour under different layer structures was similar. The anisotropy in tribological performance under different layer deposition directions was not as significant as fibre orientations.

Fig. 13 compares the SEM micrographs of worn surfaces in different printing orientations, in which the deposited layer interfaces were marked. Two types, i.e. fibre-rich, and matrix-rich regions, are characterised in 3D printed components, which is different from the uniform distribution of fibre reinforcement in conventionally manufactured composites. In particular, “bundles” of fibre inside of each filament can be clearly observed. Localised damage took place within fibre bundles, while the matrix-rich region was rather smooth without noticeable surface damages. This can be explained by stress concentration occurring in the vicinity of the fibre bundles, since fibres undertake the most load thanks to their superior mechanical properties than polymer matrix [51]. As shown in Fig. 13d, fibres were poorly jointed within the bundles as polymer matrix did not fully immerse, from where cracks were more likely to start and propagate. As discussed in the previous section, without the support from surround materials, fibres would suffer more dynamic loads/strains during the sliding process, leading to fatigue breakage of fibres. Such uneven distribution and insufficient impregnation of fibres are common issues in preparing filaments for printed composites, which limits homogeneity and mechanical performance [52,53]. On the other hand, by observing two adjacent layers, it is clear

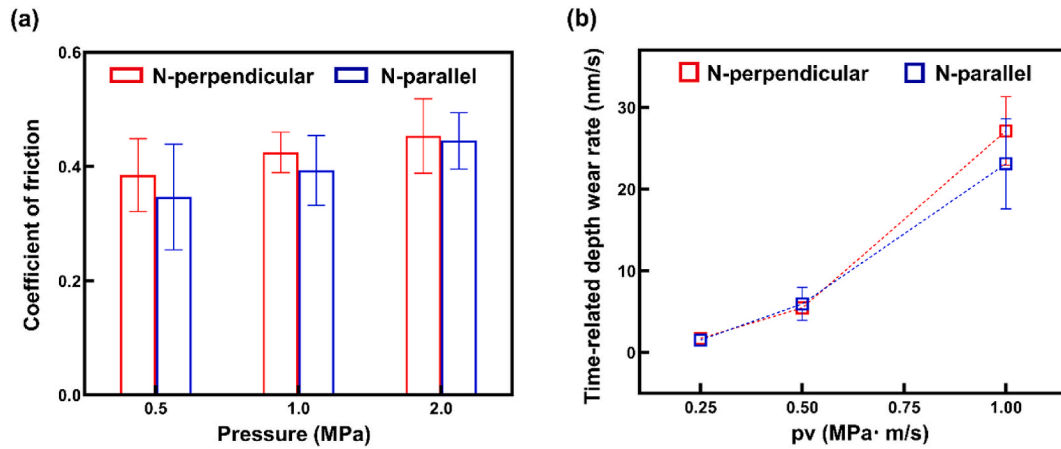


Fig. 12. Comparison of tribological performance: (a) coefficient of friction; (b) time-related depth wear rate under different printing layer directions.

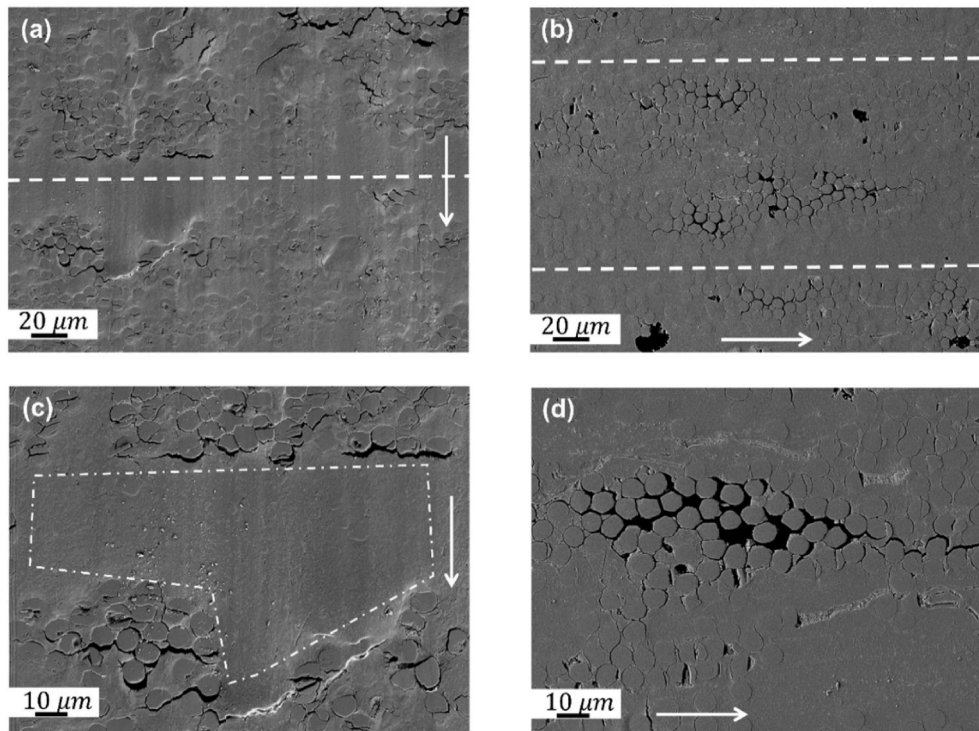


Fig. 13. Worn surfaces in different printing orientations when fibres are normally orientated: (a) perpendicular; (b) parallel against sliding direction. The arrows illustrate the sliding direction, the dash lines in (a) and (b) indicates the printing layer interface, and the dash lines in (c) distinguish the matrix-rich from fibre-rich regions.

that the interfaces between layers were well joined by polymeric matrix, as the outer parts of the filaments mainly consist of polymer rather than fibre bundles. During wear process, the cracks initiating from fibres would not pass through the interface to the neighbouring layer, i.e. the crack propagation was prohibited by the matrix material at the interface. This finding could explain the insignificant effect of printing layer direction on wear behaviour.

3.4. Effects of voids on wear behaviour

As reported in the literature, voids are commonly found in additive manufactured components due to the limited depositing pressure and incomplete intimate contact and autohesion during the 3D printing process, which will negatively impact mechanical properties [13,54]. To analyse the effect of voids on tribological performance, wear tests were

conducted on CCF/PA6 composites with CM post-processing under the same conditions as the as-printed samples. The wear results for CM process samples are summarised in Table 2.

Fig. 14 compares the friction and wear behaviour of 3D printed

Table 2

Tribological properties of CCF/PA6 composites with CM post-processing under various wear conditions.

Sliding direction	Coefficient of friction			Specific wear rate ($10^{-6} \text{ mm}^3/\text{Nm}$)		
	Pressure (MPa)			Pressure (MPa)		
	0.5	1	2	0.5	1	2
Parallel	0.46	0.55	0.57	0.64	0.78	1.83
Anti-parallel	0.49	0.52	0.51	0.44	1.30	2.08
Normal	0.34	0.38	0.37	1.91	4.8	6.73

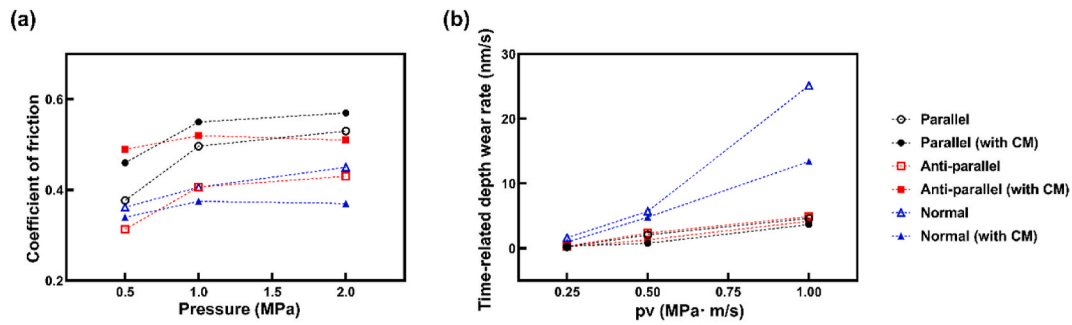


Fig. 14. Comparison between specimens with and without CM process in different fibre orientations in terms of (a) coefficient of friction, and (b) wear rate.

composites with and without CM process. As revealed in Fig. 14a, the friction under fibre *N*-orientation was reduced after the CM process. Further, the reduction became more significant with increasing applied loads. The opposite tendency can be observed when fibres were parallel and anti-parallel to the sliding direction, i.e. the samples with CM process presented higher friction coefficient than as-printed samples. The morphology of steel counterfaces showed the same characteristics as non-CM conditions, i.e. the transfer film was only formed under P and AP directions. In this case, the sliding process would be governed by like-on-like adhesion contact. Therefore, during the steady stage, the friction force would be mostly determined by the adhesion force, depending on the real contact area between two surfaces. With the reduction in voids after the CM process, the real contact area would effectively increase, leading to a higher friction force for P and AP- orientations, as shown in Fig. 14. However, the voids showed less influence on abrasion contact, especially under relatively low pressures, as the latter is more dependent on the mechanical properties, such as the hardness of the softer body of the sliding system.

Worn surfaces of samples with CM post-processing were examined and presented in Fig. 15 in order to further understand the dependence of wear mechanisms on the treatment. As shown in figures, voids were hardly seen in specimens with CM processing for all three fibre orientations. A close view of CM samples suggested a better fibre/matrix

interfacial bonding. Comparing to Figs. 7–9, fibre breakage and detachment were alleviated with the improved fibre/matrix bonding. As aforementioned, the fibre removal/breakage in *N* direction is more sensitive to fibre/matrix bonding, especially under high loading conditions. As a result, the wear resistance was effectively enhanced in *N* orientation at 2 MPa. Nevertheless, fibre-rich regions can still be easily distinguished from matrix-rich regions, as the CM processing would not affect the distribution of fibre bundles within filaments. Therefore, the basic wear mechanism has not been changed by CM processing. Thus, the overall trend was not affected, as shown in Fig. 14b.

It is interesting to note that the layer orientation or even voids showed a less dominant role, especially in *P*- and *AP*- orientations. Hence, other than the printing process, more attention may be given in future work to control the quality of filaments (such as the distribution of fibres and internal micro-sized defects). In particular, further research is necessary to enhance the bonding between polymer matrix and fillers with the proper surface treatment or using suitable coupling agents [21, 29].

4. Conclusions

In this study, tribological performance of additively manufactured CCF/PA6 composites was examined using a pin-on-disc configuration.

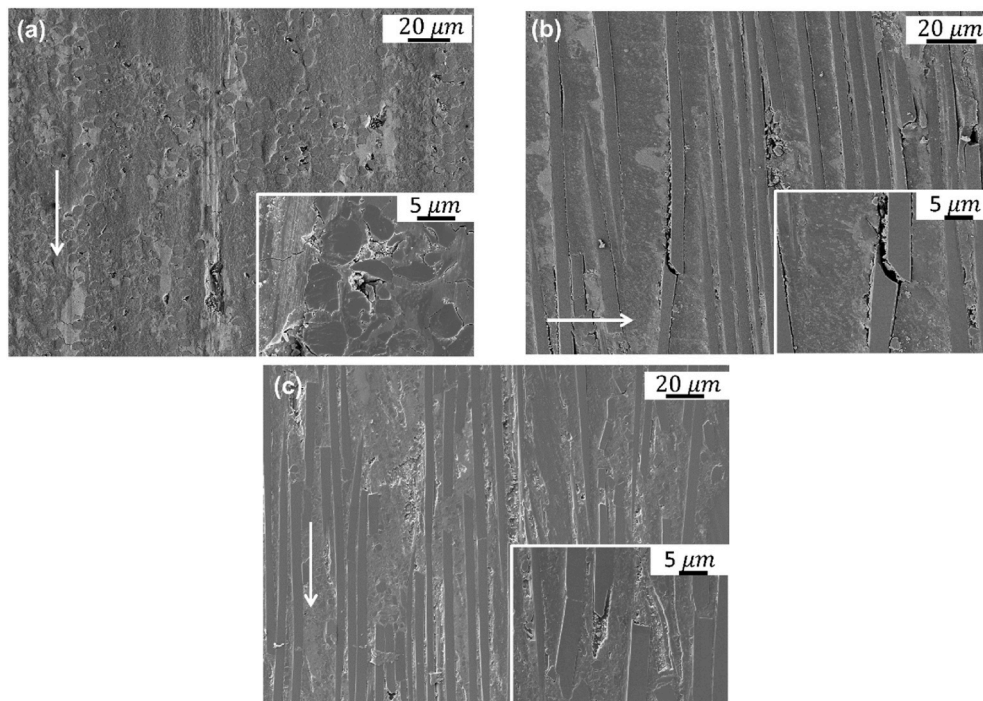


Fig. 15. Worn surfaces of CM-ed specimens in (a) *N*-; (b) *AP*-; (c) *P*- fibre orientation. The arrows indicate the sliding direction.

The effects of three fibre orientations (i.e. N-, P-, and AP-orientations), and two printing orientations (i.e. Perpendicular- and Parallel- orientations) were investigated under different loading conditions. The work explored several parameters for designing 3D printed continuous fibre reinforced polymer composites used as tribological components. It is expected that the results of the current study can illuminate favourable factors for developing high wear-resistant composites by using AM technologies. The findings can be utilised in the design of wear-resistant composite materials subjected to different sliding conditions. In particular, the following conclusions can be drawn:

1. Comparing with the printing structure such as layer orientations, fibre orientation was a predominant factor in determining the wear performance of fibre-reinforced composites. In particular, the wear process was greatly affected by fibre/matrix interfacial bonding. High stress concentration on fibres appeared noticeably in N-orientation, causing fibre/matrix debonding and fibre breakage. This finally contributed to the high wear rate of composites, especially under high loading conditions. When sliding in P and AP orientation, the fibre removal mainly occurred as fibre thinning with the enhanced wear resistance.
2. Fibre distribution and the fibre/matrix interface within the filament, were the crucial parameters affecting the overall tribological performance of the printed composite materials using the FFF method. In particular, fibre bundles were observed in worn surfaces, which would undertake most of the load during the sliding process. Accordingly, materials failure such as interface debonding and fibre breakage would more likely occur in those regions. This may explain the fact that the tribological performance of composites was little influenced by the printed structure, such as layer orientations.
3. Voids were also observed in the printed composites, which would weaken the mechanical properties of printed composites. Post-processing such as CM effectively reduced the void content and improved wear resistance and load carry capacity of composites, especially in N fibre orientation.

CRediT authorship contribution statement

Ziyan Man: Conceptualization, Data curation, Investigation, Methodology, Writing – original draft. **Hongjian Wang:** Formal analysis, Methodology, Writing – review & editing. **Qinghao He:** Formal analysis, Methodology. **Dae-Eun Kim:** Funding acquisition, Writing – review & editing. **Li Chang:** Conceptualization, Formal analysis, Funding acquisition, Methodology, Resources, Supervision, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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